

(Library Management System)

Objective - Design a Library Management System that uses inheritance to handle different types of users (e.g. Students, Professors) and Library items (e.g. Books, Magazines). The system allow borrowing & returning of items with rules varying for different user types.

Requirements:-

① Classes:-

- LibraryItem (Base class)
 - Attributes - itemId, title, isAvailable
 - Methods - borrowItem(), returnItem(), showDetails()
- Book (Derived class from LibraryItem)
 - Attributes - itemId, title, isAvailable, author
 - Methods - showDetails()
- Magazine (Derived from LibraryItem)
 - Attributes - Id, title, isAvailable, issueNumber
 - Methods - showDetails()

② User Classes:-

- User (Base class)
 - Attributes - userID, name, maxBorrowLimit
 - Methods - borrow(), returnItem(), showBorrowedItems()

- Student (Derived from User class)

Attributes - Id, name, maxBorrowedLimit = 1

Rules - can borrow only books not Magazine

Methods - borrow(), returnItem(), showBorrowedItems()

- Professor (Derived from User class)

Attributes - Id, name, maxBorrowedLimit = 1

Rules - can borrow both books & magazines

Methods - borrow(), returnItem(), showBorrowedItems()

Functional Requirements -

① Library Item :-

- The System should allow adding new library items (books, Magazines)
- The System should display detailed information about each library item (such as itemID, title, availability) when required by the user.

② Item Borrowing & Returning :-

- Users must be able to borrow available items from the library
- The system should allow users to return borrowed items, updating the availability status in the catalog
- When an item is borrowed, the system must update the user's borrowed record
- The system should enforce borrowing limit for each user type (e.g.:- students can borrow one item, Professors can borrow 1 item)

③

- Users must not be allowed more items than maximum borrowing limit

③ Item Availability Check: -

- The System must ensure that users can only borrow items that are available in the library
- If an item is already borrowed by another user, the system should prevent the current user from borrowing it

④ User Record Maintenance: -

- The System should maintain a record of borrowed items for each user
- The System should allow users to view the list of items they have borrowed at any time